

Name:		 MANIPAL UNIVERSITY JAIPUR			
Enrolment No:					
Odd Semester Makeup Examination, January 2025 Faculty of Engineering, School of Computer Science and Engineering Department of Computer Science and Engineering B.Tech.-CSE					
Course Code: CSE2103		Course: Computer Organization and Architecture	Semester: III		
Time: 3 hrs.			Max. Marks: 80		
Solution					
SECTION A					
S.No.			Marks	CO	
Q A1	What are the various factors that affect the performance of a computer system? Hardware Design Instruction Set Architecture (ISA) Compiler		[2]	CO1	
Q A2	Why does Instruction Set Architecture (ISA) require various addressing modes? To support various programming constructs.		[2]	CO2	
Q A3	Differentiate between I/O mapped and memory mapped device interface.		[2]	CO3	
	Aspect	I/O-Mapped Interface			Memory-Mapped Interface
	Addressing	Devices have a separate I/O address space, distinct from the memory address space.			Devices share the same address space as the system memory.
	Instruction Set	Access requires special I/O-specific instructions like IN and OUT (common in assembly language).			Access is done using standard memory instructions like MOV or LOAD/STORE.
	Address Space Size	Limited by the size of the I/O address bus (e.g., 8-bit or 16-bit for older systems).			Limited by the size of the memory address space, which is typically much larger.
Q A4	What is the key advantage of set-associative mapping over direct mapping? The key advantage of set-associative mapping over direct mapping is that it reduces conflict misses by allowing a memory block to map to multiple cache lines within a set, improving cache utilization and performance.		[2]	CO3	
Q A5	Write the expression for gate delay in n-bit array multiplier. 6 (n-1)-1		[2]	CO4	
SECTION B					
Q B1	A CPU has two types of instructions: Type A having 60% frequency of occurrence with a CPI of 1, and Type B having 40% frequency of occurrence with a CPI of 3. Calculate the effective CPI and execution time for 1 million instructions on a processor with a clock speed of 2 GHz.		[6]	CO1	

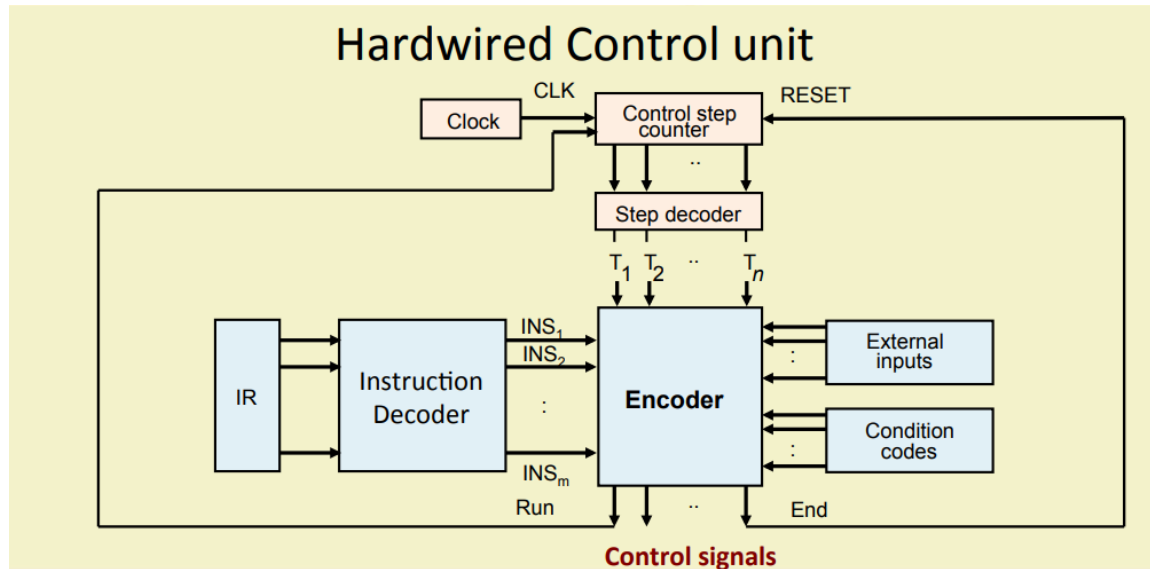
	Solution: 1. Calculate the Effective CPI: The effective CPI is the weighted average of the CPIs of the instruction types, based on their frequency of occurrence: Effective CPI = (Frequency of Type A × CPI of Type A) + (Frequency of Type B × CPI of Type B) Substitute the given values: $\text{Effective CPI} = (0.6 \times 1) + (0.4 \times 3)$ $\text{Effective CPI} = 0.6 + 1.2 = 1.8$ Answer: Effective CPI = 1.8. 2. Calculate the Execution Time: Execution time is calculated using the formula: $\text{Execution Time} = \frac{\text{Number of Instructions} \times \text{Effective CPI}}{\text{Clock Speed}}$ Substitute the given values: • Number of instructions = 1,000,000 (1 million), • Effective CPI = 1.8, • Clock speed = 2 GHz = 2×10^9 cycles/second. $\text{Execution Time} = \frac{1,000,000 \times 1.8}{2 \times 10^9}$ $\text{Execution Time} = \frac{1,800,000}{2 \times 10^9}$ Execution Time = 0.0009 seconds = 0.9 milliseconds. Answer: Execution time = 0.9 milliseconds. 3 marks for effective CPI and 3 marks for Execution time.		
Q B2	Why didn't processor manufacturers increase clock speeds instead of making them multicore? Describe various processor designs according to Flynn's Classification. • It is difficult to sustain Moore's law and at the same time meet performance demands of various applications. – Difficult to increase clock frequency, mainly due to power consumption issues. • Possible solution: – Replicate hardware and run them at a lower clock rate to reduce power consumption. – 1 core running at 3 GHz has the same performance as 2 cores running at 1.5 GHz, with lower power consumption.	[6]	CO1
	3 marks		

	• Single instruction-stream single data-stream (SISD) – Traditional uniprocessor systems. • Multiple instruction-stream single data-stream (MISD) – No commercial implementation exists. – Pipelining can be argued as a type of MISD processing. • Single instruction-stream multiple data-stream (SIMD) – Array and vector processors. • Multiple instruction-stream multiple data-stream (MIMD) – Multiprocessor systems (various architectures exist). 3 marks		
Q B3	Write an assembly language sequence for the following 'C construct' given below. <pre> if ((x > y) && (x > z)) { count ++; CMP y, x Branch < 0 Label CMP z, x Branch < 0 Label ADD #1, R8 Label: </pre> 2 marks if CMP instruction is used, 4 marks if both conditions are checked.	[6]	CO2
Q B4	What are the two modes of DMA data transfer, and how do they differ in operation? DMA transfer can take place in two modes: a) DMA cycle stealing • The DMA controller requests for the for a few cycles 1 or 2. • Preferably when the CPU is not using memory. • DMA controller is said to steal cycles from the CPU without the CPU knowing it. b) DMA block transfer • The DMA controller transfers the whole block of data without interruption. • Results in maximum possible data transfer rate. • CPU will lie idle during this period as it cannot fetch any instructions from memory.	[6]	CO3
Q B5	A system features a 32 KB cache with each block being 64 bytes. The main memory size is 1 MB. Determine the following: - The total number of cache lines. - The total number of blocks in main memory. - The specific cache line to which the block at the memory address AA3F5 (in hexadecimal) maps.	[6]	CO3

	Tag offset=5bit Line offset=9bit Block Offset=6bit Final Answers: (i) 512 (ii) 16,384 (iii) 143 {2+2+2}																													
SECTION-C																														
Q C1	Starting from memory location LIST, 100 alphabets are stored (Some in lowercase and some are in uppercase). Consider the following assembly code to convert the lowercase alphabets to uppercase and uppercase alphabets will remain intact. MOV #LIST, R0 MOV #100, R1 LOOP CMP (R0), #97 BRANCH > 0 NEXT SUB #32, (R0) NEXT INC R0 DEC R1 BRANCH > 0 LOOP The above code is executed on a processor having a clock rate of 1 GHz. The processor can execute an instruction that access memory for reading or writing data with an average of 5 steps, and other instructions that do not require operand access from or to main memory can execute with an average of 3 steps (each step completes in one clock cycle). Determine how long the above assembly code will take to execute, given that 50 of the alphabets are lowercase and 50 are uppercase. <table><tr><td></td><td></td><td>$N \times S$</td></tr><tr><td></td><td>MOV #LIST, R0</td><td>1×3</td></tr><tr><td></td><td>MOV #100, R1</td><td>1×3</td></tr><tr><td>LOOP</td><td>CMP (R0), #97</td><td>100×5</td></tr><tr><td></td><td>BRANCH > 0 NEXT</td><td>100×3</td></tr><tr><td></td><td>SUB #32, (R0)</td><td>50×5</td></tr><tr><td>NEXT</td><td>INC R0</td><td>100×3</td></tr><tr><td></td><td>DEC R1</td><td>100×3</td></tr><tr><td></td><td>BRANCH > 0 LOOP</td><td>100×3</td></tr></table> T = N×S / R = 1956 * 1 ns T = 1.956 microseconds 2 marks for N*S attempt +3 marks for correct N*S			$N \times S$		MOV #LIST, R0	1×3		MOV #100, R1	1×3	LOOP	CMP (R0), #97	100×5		BRANCH > 0 NEXT	100×3		SUB #32, (R0)	50×5	NEXT	INC R0	100×3		DEC R1	100×3		BRANCH > 0 LOOP	100×3	[10]	CO2
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+2 marks for expression

Design a hardwired control unit for a processor that supports up to 128 instructions in its Instruction Set Architecture (ISA), with control sequences spanning a maximum of 7 timesteps.



[4 marks]

Q C2

Specify the sizes of the instruction decoder, control step counter, and step decoder used in the design.

[10]

CO5

Instruction Decoder size is 7-to-128-line decoder

Control step counter is a 3-bit counter

The control step decoder size is 3-to-8-line decoder

[2 marks]

Additionally, write the control sequences *ADD R1, R2, R3*; *Move R1, R2* and *BRANCH LABEL* considering three bus processor datapath.

T1: PC_{out} , $R = B$, MAR_{in} , READ, IncPC

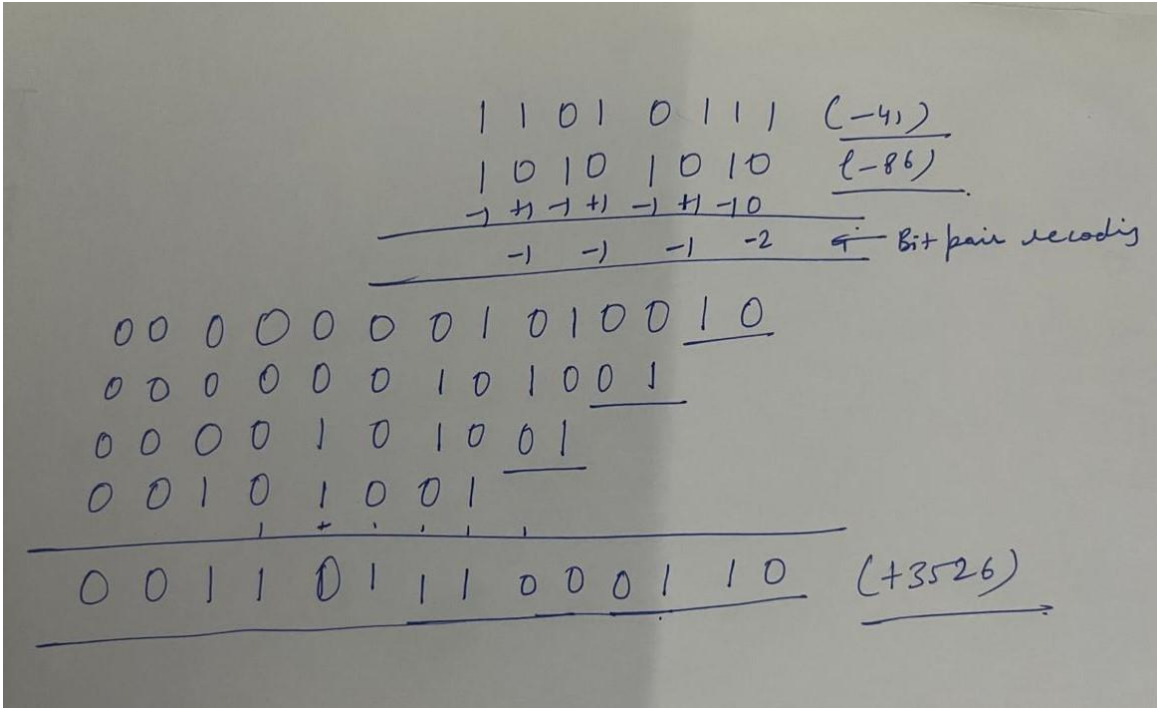
T2: WMFC

T3: MDR_{outB} , $R = B$, IR_{in}

[2 marks for first three common steps]

	For ADD R1, R2, R3 T4: R1outA, R2outB, Select A, ADD, R3in, End For Move R1, R2 T4: R1outB, R=B, R2in, End For Branch Label T4: Offset field of IRout, PCout, Select A, ADD, PCin, End 2 marks		
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Section-D

Q D1	Apply the bit-pair recoding algorithm to multiply the 8-bit 2's complement numbers A = 11010111 (multiplicand) and B = 10101010 (multiplier). Additionally, calculate the gate delays required to compute the product if the summands are added using a 16-bit Carry Look-Ahead (CLA) adder constructed from cascading 4-bit CLA adders.  For 16-bit Carry Look-Ahead (CLA) adder constructed from cascading 4-bit CLA adders, gate delay for C16 and S15 are 9 and 10 respectively. Here, 4 summands are there so three additions require a total of 30 gate delays. Correct Bit pair recoding: 2 marks Correct Summands: 2 marks Correct answer: 2 marks Correct GD for 16-bit adder: 2 marks 4 summands are there so three additions require a total of 30 gate delays: 2 marks	[10]	CO4
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QD2 A pipelined processor consists of five stages: Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Write Back (WB), with each stage taking 2 ns. There is an additional overhead of 0.5 ns per stage due to control logic. Given the following conditions:

1. **Ideal Case:** Calculate the total time required to execute 100 instructions without any stalls. [3 marks]
2. **With Hazards:** Assume 10% of the instructions encounter a 2-cycle stall due to data hazards. Determine the new execution time. [3 marks]
3. **Speedup:** Compare the execution time of the pipelined processor (with and without hazards) to that of a non-pipelined processor, which executes instructions sequentially at 10 ns per instruction. [2 + 2 marks]

Solution:

1. **Ideal Case:**

- Total time per stage = $2 + 0.5 = 2.5$ ns.
- For the first instruction: $5 \times 2.5 = 12.5$ ns.
- For the remaining 99 instructions, each takes 1 stage = 2.5 ns.

Total time:

$$\text{Total time} = 12.5 + (100 - 1) \times 2.5 = 12.5 + 247.5 = 260 \text{ ns.}$$

Answer: Total time in ideal case = 260 ns.

2. **With Hazards:**

- 10% of the instructions encounter a 2-cycle stall.
- Number of stalled instructions = $100 \times 0.1 = 10$.
- Each stall adds $2 \times 2.5 = 5$ ns.

Additional stall time:

$$\text{Total stall time} = 10 \times 5 = 50 \text{ ns.}$$

New total execution time:

$$\text{New total time} = 260 + 50 = 310 \text{ ns.}$$

Answer: Total time with hazards = 310 ns.

3. Speedup:

- **Non-Pipelined Processor:**

Each instruction takes 10 ns to execute sequentially.

Total time for 100 instructions:

$$\text{Total time (non-pipelined)} = 100 \times 10 = 1000 \text{ ns.}$$

- **Speedup Without Hazards:**

$$\text{Speedup} = \frac{\text{Time (non-pipelined)}}{\text{Time (pipelined ideal)}} = \frac{1000}{260} \approx 3.85.$$

- **Speedup With Hazards:**

$$\text{Speedup} = \frac{\text{Time (non-pipelined)}}{\text{Time (pipelined with hazards)}} = \frac{1000}{310} \approx 3.23.$$

Answer:

- Speedup without hazards = **3.85**.
- Speedup with hazards = **3.23**.